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Developmental Mechanisms of Topographic Map Formation and Alignment

Jianhua Cang¹ and David A. Feldheim²

¹Department of Neurobiology, Northwestern University, Evanston, Illinois 60208; email: cang@northwestern.edu

²Department of Molecular, Cell and Developmental Biology, University of California, Santa Cruz, California 95064; email: dfeldhei@ucsc.edu

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Abstract

Brain connections are organized into topographic maps that are precisely aligned both within and across modalities. This alignment facilitates coherent integration of different categories of sensory inputs and allows for proper sensorimotor transformations. Topographic maps are established and aligned by multistep processes during development, including interactions of molecular guidance cues expressed in gradients; spontaneous activity-dependent axonal and dendritic remodeling; and sensory-evoked plasticity driven by experience. By focusing on the superior colliculus, a major site of topographic map alignment for different sensory modalities, this review summarizes current understanding of topographic map development in the mammalian visual system and highlights recent advances in map alignment studies. A major goal looking forward is to reveal the molecular and synaptic mechanisms underlying map alignment and to understand the physiological and behavioral consequences when these mechanisms are disrupted at various scales.

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INTRODUCTION

The brain detects and processes sensory information using separate, parallel pathways and then combines them to create a unified percept of the outside world and to issue appropriate motor commands. Parallel pathways include those from different sensory modalities, those from bilateral sensory organs within the same modality, and those that signal different aspects of the same stimulus. The ability to integrate multiple channels of information in an organized way improves the sensitivity to detect stimuli and solves ambiguities that arise from

the processing of a single attribute (Stein & Meredith 1993). For example, viewing through both eyes and hearing with both ears greatly improve our ability to perceive depth and to localize sound sources. Sensory integration is not only beneficial, but also necessary for our proper perception and interaction with the environment. Failure to integrate different sensory modalities may contribute to a number of neurological deficits, including autism spectrum disorders, synesthesia, dyslexia, and many psychoses (Bargary & Mitchell 2008, Brock et al. 2002, Geschwind & Levitt 2007).

Sensory integration:
when neuronal circuits
combine inputs from
multiple sensory
modalities to generate
meaningful outputs

Thus two fundamental goals of neuroscience research are to understand the developmental mechanisms that create the brain circuitry capable of accurate multichannel integration and to understand the physiological and behavioral consequences of disrupting these mechanisms.

Topographic mapping is a feature of brain organization that may facilitate or even be required for proper sensory integration and sensorimotor transformations. In the brain, neurons are often organized into maps according to their functional properties, such as where in space they respond and the stimulus features to which they are tuned. These maps have stereotypical organization: Neighboring neurons share similar response properties, and the value of the mapped attribute changes progressively. Some topographic maps arise from preserving the spatial order of the receptor neurons in the sensory epithelium. These include retinotopic maps of visual space, tonotopic maps of sound frequency, and somatotopic maps of body surface. In these maps, neuronal neighbor-to-neighbor relationships in the projecting structure are maintained in their targets, thus preserving a continuous representation in many brain areas (Luo & Flanagan 2007). Other maps are generated after the sensory information is transformed by neuronal computations (Knudsen et al. 1987). The value of the computed parameter, instead of the input's spatial relationship, is then mapped systematically. These include maps of sound location in the auditory system and maps of orientation preference in the visual cortex (Hubel et al. 1977, Knudsen & Konishi 1978a). These computational maps can then be preserved in successive brain areas by topographically organized projections. Both types of topographic maps also exist in the motor systems: Neurons may be organized according to which body areas they control or the movement amplitude they command (Penfield & Boldrey 1937, Wurtz & Goldberg 1972).

The orderly organization of sensory and motor maps offers a structural substrate for proper multichannel integration and sensori-

motor transformation. For example, to integrate the sight and sound of an object, visual and auditory neurons that respond to the same locations need to be connected or converge on the same target. Aligning the visual and auditory space maps naturally puts these neurons near each other, thus simplifying the connectivity pattern between the two maps. Theoretically, individual neurons in the two systems could still be connected according to their receptive field locations even in the absence of topographic maps, but such a connectivity pattern would be extremely difficult to establish during development and costly in terms of required wiring (Chklovskii & Koulakov 2004).

The past decade or so has seen an explosion in knowledge of the molecular and activity-dependent mechanisms that are used to form topographic maps, especially within the mammalian visual system. These studies have provided a theoretical framework, and perhaps more importantly, genetic models, that can now be used to understand how multiple maps become aligned and the resulting behavioral consequences when the alignment is disrupted. In this review, we first summarize our current understanding of topographic map development in the mammalian visual system and then review recent advances in map alignment studies. In particular, we focus most of our discussion on the superior colliculus (SC) and its nonmammalian homolog, the optic tectum (OT).

The Superior Colliculus: An Integrative Sensorimotor Center with Aligned Topographic Maps

In the mammalian visual system, the retinal ganglion cells (RGCs) send projections to several targets involved in image- and non-image-forming visual behaviors. Two of the main targets are the SC in the midbrain and the dorsal lateral geniculate nucleus (dLGN) in the thalamus. The dLGN, in turn, projects to V1 in the posterior cerebral cortex, which subsequently projects to several higher visual areas responsible for conscious vision and multimodal integration. Each component of the visual system

Sensorimotor transformation: the process that the brain converts sensory stimuli into motor commands

Receptive field: the portion of sensory space that can elicit neuronal responses when stimulated

SC: superior colliculus

OT: optic tectum

RGC: retinal ganglion cell

V1: primary visual cortex

dLGN: dorsal lateral geniculate nucleus

Multimodal integration: when neuronal circuits combine inputs from multiple sensory modalities to generate meaningful outputs

A-P: anterior-posterior
M-L: medial-lateral
N-T: nasal-temporal
D-V: dorsal-ventral
S1: primary somatosensory cortex

is mapped topographically (Andermann et al. 2011, Lewin 1994, Marshel et al. 2011, Wang & Burkhalter 2007). Because of its importance in multimodal sensorimotor transformation and experimental accessibility, the SC/OT has been a leading model to study how topographic maps form and align during development. The architecture of the SC is conserved among mammals and has two key features: lamination along the superficial to deep axis, which organizes inputs according to where they originate; and topography along the anterior-posterior (A-P) axis (also called the rostral-caudal axis) and the medial-lateral (M-L) axis, which gives spatial identity to a stimulus (**Figure 1**).

The superficial to deep axis of the SC contains seven alternating fibrous and cellular laminae, which can be divided into superficial and deep layers (Gandhi & Katnani 2011, Stein 1984). The most superficial cellular lamina of the SC, the stratum griseum superficiale (SGS), receives RGC input from both eyes and is also laminated with different RGC types that terminate in different sublaminae within the SGS (Inoue & Sanes 1997). The SC also receives visual input from V1 that terminates in the lower SGS (Wang & Burkhalter 2013). These three sources of visual inputs are mapped topographically with respect to the visual field and are aligned in retinotopic register with each other. The nasal-temporal (N-T) axis of the visual field (azimuth) is represented along the A-P axis of the SC and the dorsal-ventral (D-V) axis (elevation) represented along the M-L axis of the SC (**Figure 1**).

The deep layers of the SC receive inputs from the primary somatosensory cortex (S1), the auditory nuclei of the midbrain, the trigeminal nucleus of the brain stem, and many other sensory and motor areas (May 2005). The somatosensory and auditory inputs are organized into topographic maps that are aligned with the visual maps in the superficial layers (**Figure 1**) such that neurons in the same vertical column have corresponding receptive field locations, either in visual and auditory space or on the body surface (Drager & Hubel 1975, Stein 1984). Multisensory integration takes place in the SC

as many neurons in the deep layers respond to more than one modality owing to topographically organized projections from the superficial to deep layers. Finally, these sensory maps are also aligned with motor maps in the deep layers of the SC that encode the direction and amplitude of saccadic eye and head movements (Sparks et al. 1990, Wurtz & Albano 1980). The alignment of these maps allows the SC to integrate visual, auditory, and tactile information and to initiate orienting movements to redirect attention toward a stimulus.

MOLECULAR AND ACTIVITY-DEPENDENT MECHANISMS ARE USED TO CREATE TOPOGRAPHIC MAPS

Investigators have traditionally proposed two opposing mechanisms for topographic mapping, which apply to the development of neural circuits in general (Cline 2003). One is a genetic model whereby the generation of precise neural circuitry results from specific gene expression, which gives each neuron instructions about where to project and with whom to synapse. The other is an activity-dependent model, which postulates that the firing patterns of a neuron's action potentials are used to instruct map creation. Not surprisingly, our current understanding is that the development of topographic maps uses a combination of both mechanisms. Because the mechanistic details of map formation have been extensively reviewed in recent years (Feldheim & O'Leary 2012, Flanagan 2006, McLaughlin & O'Leary 2005, Suetterlin et al. 2012, Triplett & Feldheim 2012), this section is limited to the general principles of topographic mapping that are relevant to understanding map alignment.

Genetic Mechanisms of Retinocollicular Map Formation

The initial insights that led to genetic models of map development came from studies in lower vertebrates, where retinal axons regenerate after optic nerve severing and reconnect to the

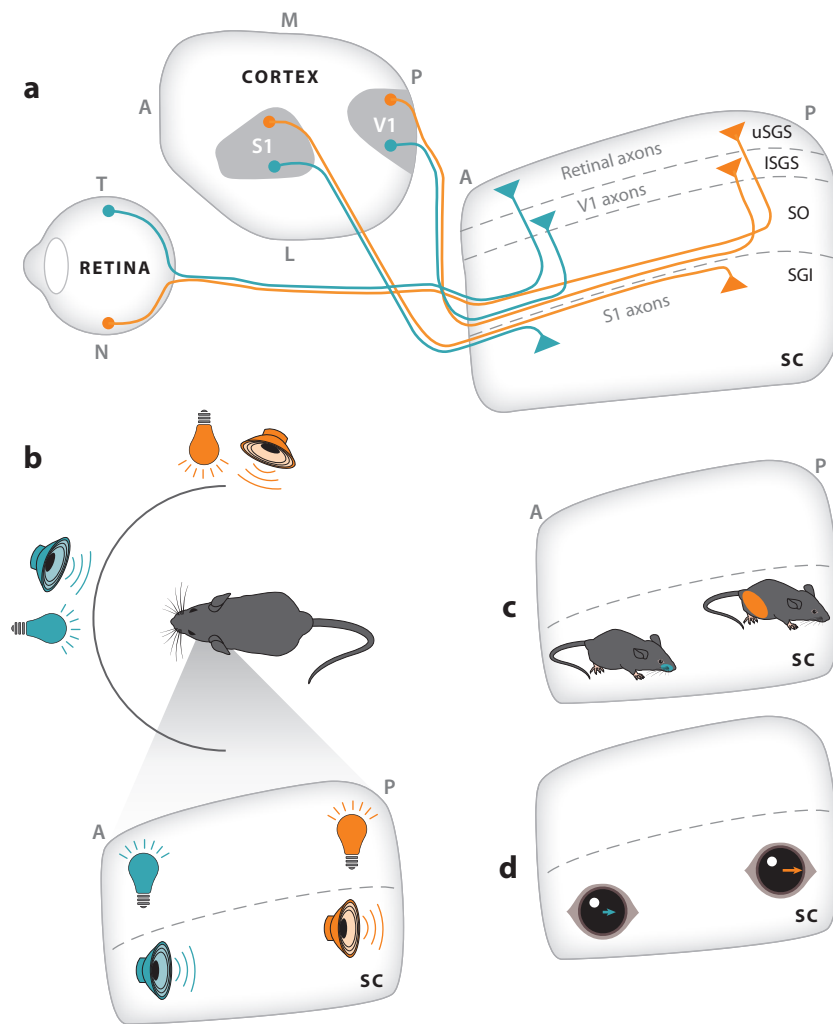


Figure 1

Alignment of multiple maps in the superior colliculus (SC). (a) A schematic diagram showing the laminated nature of the SC and inputs from the retina, the visual cortex (V1), and the somatosensory cortex (S1). Each projection is mapped topographically and aligned with the others. Abbreviations: N, nasal; T, temporal; A, anterior; P, posterior; L, lateral; M, medial; uSGS, upper stratum griseum superficiale; ISGS, lower stratum griseum superficiale; SO, stratum opticum; SGI, stratum griseum intermediale. (b–c) The orderly projections result in aligned sensory maps in the SC such that neurons in a vertical column have corresponding receptive field locations, either in visual and auditory space (b) or on the body surface (c), as illustrated by the same color in the diagrams. (d) Stimulation of deep layers of the anterior SC evokes small eye movements, whereas posterior SC stimulation evokes larger movements, consistent with the sensory map orientation.

OT in the same patterns as they did during development. If the eyes were rotated 180°, the regenerating axons still grew back to their original destination rather than reorienting to

maintain the layout of the visual field, even though this led to maladaptive behaviors in response to visual stimuli (Sperry 1943). These results led Sperry to propose the chemoaffinity

hypothesis. Sperry predicted that genetically encoded labels exist in both the retina and the OT; furthermore, if these labels are expressed in continuous gradients across both structures, each RGC axon and target neuron would have a unique concentration of labels that could be matched up to establish topography (Sperry 1963). Bonhoeffer and colleagues later demonstrated that these labels were cell surface proteins using an *in vitro* stripe assay that recapitulated some key aspects of retinotectal mapping along the N-T axis (Walter et al. 1987a). In this assay, temporal or nasal retinal explants were presented with alternating stripes of membranes derived from the anterior or posterior chick OT. The temporal axons avoided membranes from the posterior OT, whereas nasal axons showed no preference for either substrate. Treatment of posterior membranes with either a protease or PI-phospholipase C (PI-PLC) abolished the preference of temporal axons for anterior membranes. This result suggested that a glycosylphosphatidylinositol (GPI)-linked cell surface protein that was enriched in the posterior OT and triggered repulsion was responsible for the choice of temporal axons (Walter et al. 1987a,b).

Identification of EphAs and Ephrin-As

Two different approaches led to the discovery of the Eph and ephrin family of proteins whose properties were predicted by Sperry and Bonhoeffer (for reviews see Drescher et al. 1997, Flanagan & Vanderhaeghen 1998); we now know that Eph/ephrin signaling molecules are found in all multicellular species but have been greatly expanded in mammals (Chisholm & Tessier-Lavigne 1999, Mellott & Burke 2008). Ephs are receptor tyrosine kinases (RTKs) that bind to ephrins. Among the major RTK/ligand families, Eph/ephrin signaling is distinctive because both ephrins and Ephs are membrane bound, resulting in signals that are localized to areas of cell contact. After binding, both Eph- and ephrin-expressing cells can transduce signals that lead to changes in cellular behavior. Thus, Ephs and ephrins signal bidirectionally:

ephrin induction of Eph signaling is termed forward signaling, and Eph induction of ephrin signaling is called reverse signaling. Bidirectional signaling facilitates the regulation of axon and target cell responses concomitantly after an interaction (Kullander & Klein 2002).

Ephs and ephrins can each be subdivided into two classes. Ephrin-As are GPI-linked to the membrane, whereas ephrin-Bs are transmembrane proteins that have cytoplasmic domains with C-terminal PDZ binding motifs. EphAs and EphBs are distinguished by sequence similarities that dictate their binding specificity for either ephrin-As or -Bs. Binding partners are promiscuous *in vitro* within each subfamily; each EphA can bind multiple ephrin-As, and each EphB can bind multiple ephrin-Bs. Some crosstalk also exists between subfamilies. For example, EphA4 can bind all ephrin-Bs, and EphB2 can bind to ephrin-A5 (Gale et al. 1996, Himanen et al. 2004). Although discussed here in the context of topographic mapping, Ephs and ephrins are important in a multitude of cellular and developmental events and can act as oncogenes and tumor suppressors when mutated or misregulated (Holmberg et al. 2005, 2006; Miao & Wang 2012; Pasquale 2010).

Both Forward Signaling and Reverse Signaling Build Topographic Maps Along the N-T Mapping Axis

In all species that researchers have examined, including humans, multiple EphA and ephrin-A family members are expressed in complementary gradients along the N-T axis of the retina and along the N-T mapping axis in the SC/OT (Lambot et al. 2005). Multiple EphAs are expressed in a high-temporal to low-nasal gradient in the retina and in a high-anterior to low-posterior gradient in the SC. Ephrin-As are expressed in a complementary manner, having a high-nasal to low-temporal expression in the retina and a low-anterior to high-posterior expression pattern in the OT (**Figure 2**). These counter gradients, together with the finding that ephrin-A-bearing axons are repelled by EphAs (Rashid et al. 2005), help

explain topographic map formation. Temporal RGC axons have high levels of EphA receptors, are thus easily repelled by collicular derived ephrin-As, and therefore map to the anterior SC/OT, where ephrin-A expression is lowest (forward signaling). Nasal axons contain high levels of ephrin-As, are repelled by EphAs in the anterior SC, and therefore map to the posterior SC, where repulsion is lowest (reverse signaling). Axons will terminate in a region of the SC/OT where their sensitivity to the two repellent activities is minimized (Nakamoto et al. 1996, O'Leary & McLaughlin 2005, Rashid et al. 2005). Consistent with this idea, perturbation of EphA and ephrin-A levels in the retina or in the SC disrupts topographic mapping, which can be revealed by both anatomical and functional assays (**Figure 3**) (Cang et al. 2008b; Carreres et al. 2011; Feldheim et al. 2000, 2004; Frisén et al. 1998; Rashid et al. 2005).

One benefit of having counter gradients of Ephs and ephrins is that the same labels can be used repeatedly at successive stages within the same pathway. EphA and ephrin-A counter gradients are found in each visual area and guide the maps from the retina to the SC and the dLGN, from the dLGN to V1, and V1 projections to the dLGN and the SC (Cang et al. 2005a, Feldheim et al. 1998, Pfeiffenberger et al. 2006, Torii & Levitt 2005, Triplett et al. 2009) (**Figure 2**). In addition, EphA/ephrin-A signaling is required for the topographic mapping of many other projections, including those of the somatosensory, motor, auditory, hippocampal, and olfactory systems (Cramer 2005, Cutforth et al. 2003, Galimberti et al. 2010, Serizawa et al. 2006, Vanderhaeghen et al. 2000).

Relative Levels of EphAs Are Used to Map RGC Axons Topographically

An important insight into how EphA/ephrin-A gradients direct topographic mapping comes from experiments by Lemke and colleagues, which demonstrated that topography is established through relative, as opposed to absolute, levels of EphA signaling (Bevins et al. 2011, Brown et al. 2000, Reber et al. 2004). Taking

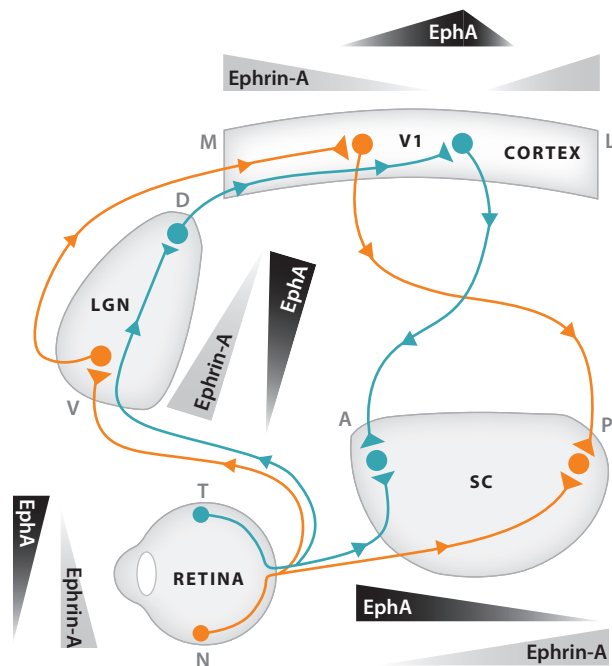


Figure 2

EphA and ephrin-A counter gradients are found in each area of the developing visual system, including the retina, the superior colliculus (SC), the lateral geniculate nucleus (LGN), and the primary visual cortex (V1). Abbreviations: N, nasal; T, temporal; A, anterior; P, posterior; D, dorsal; V, ventral; L, lateral; M, medial.

advantage of the fact that EphA3 is not endogenously expressed in the mouse retina, Brown et al. (2000) introduced EphA3 into the 3' untranslated region of the Islet-2 locus. Because Islet-2 is expressed in approximately half of all RGCs distributed relatively evenly across the retina, two populations of RGCs exist: (a) Islet-2⁻ cells, which express endogenous levels of EphA4, EphA5, and EphA6 according to their positions along the T-N axis; and (b) Islet-2⁺ cells, which express exogenous EphA3 in high amounts in addition to the endogenous EphAs gradients. As a result, the RGCs expressing EphA3 are compressed into the anterior half of the SC, whereas the Islet-2⁻ RGCs, although having wild-type (WT) levels of EphA, are restricted to the posterior half of the SC (Brown et al. 2000). This anatomical phenotype results in two complete and functional representations of the visual fields as revealed by intrinsic imaging (Triplett et al. 2009).

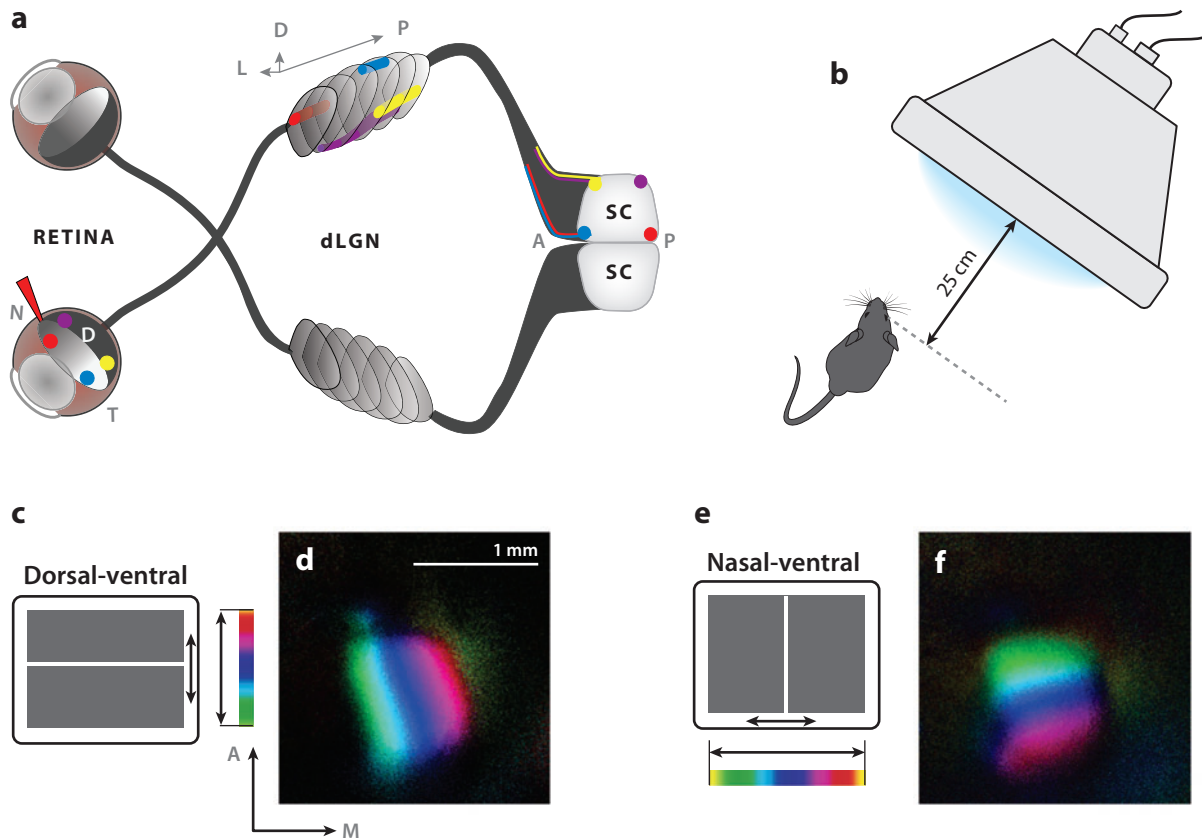


Figure 3

Determining retinotopic maps using anatomical tracing and functional imaging. (a) Anatomical tracing of topographic maps. Focal injections of the lipophilic fluorescent DiI in the retina label RGC axons and the termination zones in their targets. Shown is a summary of the topography of the retinogeniculate and retinocollicular projections in wild-type mice. (b–f) Functional maps in the SC revealed by optical imaging of intrinsic signals. An anesthetized mouse is positioned to view a computer monitor through the eye contralateral to the hemisphere under study (b). Stimuli are thin moving bars (c and e). The elevation map of a WT mouse is shown in panel d; the retinotopy is represented according to the color scale shown in panel c, and the response magnitude is illustrated by brightness. The azimuth map of the same SC is shown in panel f. Abbreviations: N, nasal; T, temporal; A, anterior; P, posterior; D, dorsal; L, lateral; M, medial; dLGN, dorsal lateral geniculate nucleus.

Competition for Target Space in Topographic Mapping

How do axons read relative rather than absolute levels of ephrins? Axon-axon competition for some limiting factor(s) in the target may be important (Brown et al. 2000, Koulakov & Tsigankov 2004, Prestige & Willshaw 1975, Triplett et al. 2011, Tsigankov & Koulakov 2006). This limiting factor would attract axons, which could provide a force that counterbalances the repellent gradient of ephrin-As.

Competition for this factor would tend to drive axons with lower EphA expression in a posterior direction. Competition models also explain a large body of work involving ablations of parts of the retina or OT/SC. These experiments indicated that, at least under some experimental conditions, retinal axons tend to spread out and fill the available space in the target (Finlay et al. 1979, Fraser & Hunt 1980, Simon et al. 1994).

Experiments in fish and mice have tested the role of axonal competition in topographic

mapping. Gosse et al. (2008) created transgenic zebrafish that contained only one RGC and assayed its projection patterns in the OT. Competition-based models predicted that the RGC's termination would shift anteriorly because positive factors were available in the anterior SC/OT. Instead, the most distal branch of the axon projected to a position similar to what it would have under the crowded conditions in a WT retina (Gosse et al. 2008). Experiments in mice gave an opposite result. In *Math5* (*Atoh7*) mutant mice, which have severely reduced numbers of RGCs (Brown et al. 2001), the RGC axons do not fill the entire SC but instead are enriched in the antero-medial portion. These data suggest that in the absence of competition all RGCs prefer to project to where repulsion from ephrin-A ligands is minimized. Lower vertebrates may use competition-based mechanisms to a lesser degree than mammals, which could be due to the difference in how RGCs find their appropriate termination zone (TZ) position in fish and mammals (O'Leary & McLaughlin 2005). In fish, the RGC axon extends directly to the correct location. In contrast, mouse RGCs overshoot their final locations and then, over a period of a few days, form TZs by interstitial branching, which may be influenced more easily by competition.

The competition in mice could be mediated by comparing axonal EphA levels, which would result in the spatial segregation of retinal axons in the SC (Reber et al. 2004). Alternatively, a recent computational study (Grimbert & Cang 2012) suggested that the competition could instead rely on the optimization of collicular resources mediated by neurotrophic receptors such as p75, in addition to p75's role in reverse signaling as a transmembrane signaling partner for ephrin-As (Lim et al. 2008).

Mapping the Dorsal-Ventral Axis of the Retina onto the Superior Colliculus

An important result from the gain- and loss-of-function experiments that manipulate EphA and ephrin-As is that the topographic mapping

defects are restricted predominantly to the N-T mapping axis, leaving the D-V mapping axis intact. This result strongly indicates that each axis uses independent mechanisms for mapping (Cang et al. 2008a). Although the mechanisms of D-V mapping are not as well understood, investigators have proposed that a combination of pretarget sorting and graded signaling by EphB/ephrin-Bs is used to sort axons along the D-V mapping axis of the SC.

As RGC axons exit the optic disc at the back of the eye, they tightly fasciculate to form the optic nerve and join axons from the opposite eye once they reach the optic chiasm at the base of the brain. They subsequently defasciculate prior to entering their targets, including the SC. Experiments in which the dorsal or ventral retina is labeled with DiI reveal that D-V order is established within the optic tract before RGC axons enter the SC (Plas et al. 2005, Simon & O'Leary 1991). Mutations that affect D-V patterning in the retina also affect D-V sorting and mapping in the SC, suggesting that optic tract sorting is important for topographic mapping (Plas et al. 2005). Although the molecular mechanisms regulating RGC axon sorting are unknown, similar tract organization in the olfactory system relies on Sema3a and its receptor, neuropilin-1 (Bozza et al. 2009, Imai et al. 2009). On the other hand, RGC axons also refine their M-L position within the SC by interstitial branching off their primary axon toward their appropriate position. These branches can form either medial or lateral to the primary axon, suggesting that the exact entrance location of the axon into the SC is not critical (Hindges et al. 2002, McLaughlin et al. 2003a, Nakamura & O'Leary 1989).

The expression patterns and in vivo loss-and gain-of-function experiments have led researchers to propose that both EphB/ephrin-B forward and reverse signaling regulate D-V topographic map formation in the SC/OT (Hindges et al. 2002, Mann et al. 2002, Thakar et al. 2011). Dual gradients of EphBs and ephrin-Bs are expressed within the retina; Ephrin-Bs are expressed in a high-dorsal to low-ventral gradient, whereas EphBs are

Optic chiasm:

located at the bottom of the brain; where the optic nerves partially cross

expressed in the opposite, high-ventral to low-dorsal gradient (Braisted et al. 1997, Hindges et al. 2002, Holash & Pasquale 1995, Mann et al. 2002, Marcus et al. 1996). However, the expression patterns of EphBs and ephrin-Bs are not as obviously graded in the M-L axis of the SC/OT. In the mouse, mRNA in situ hybridization reveals ephrin-B1 in a high medial to low lateral gradient restricted to the ventricular zone of the SC, far from the superficial retinorecipient layers. A similar pattern of expression exists in the chick OT, where ephrin-B1 protein expression is associated with radial glia that span the depth of the OT, suggesting expression on these cells could guide incoming retinal axons (Braisted et al. 1997, Hindges et al. 2002). EphBs are widely expressed in the vertebrate tectum, but not in an obvious medial-lateral gradient, making it hard to understand how EphB/ephrin-B interactions could generate positional identity on their own (Higenell et al. 2011, Scalia & Feldheim 2005, Thakar et al. 2011). Despite the lack of obvious gradients in the SC/OT, EphB/ephrin-B mutants do display topographic mapping defects. The retinocollicular maps have been analyzed in mouse lines that remove all or partial functions of EphB/ephrin-B signaling (Hindges et al. 2002, Thakar et al. 2011). In each case, many RGCs make M-L and A-P mapping errors, but most RGCs do map correctly, indicating that other D-V mapping molecules are still to be identified.

Other Molecules in Mapping Retinocollicular Projection

Along the N-T mapping axis, the GPI-linked protein repulsive guidance molecule (RGM) and its receptor neogenin are expressed in gradients along the N-T mapping axis. RGM can repel temporal but not nasal axons in vitro (Monnier et al. 2002, Rajagopalan et al. 2004). Perturbation of RGM levels by overexpression or by using RNAi disturbs the retinal-tectal projection in chicks (Monnier et al. 2002, Rajagopalan et al. 2004). Multiple semaphorins and their receptors plexins and neuropilins are expressed in the retina and the OT/SC during

development, and *Sema3A* can repel *Xenopus* retinal axons in vitro (Campbell et al. 2001).

Molecules that have been implicated in D-V mapping include Wnts and their receptors and the cell adhesion molecules L1 and ALCAM. Wnt3 is expressed in a high to low M-L gradient in chick OT and mouse SC, and members of the Ryk and Frizzled families of Wnt receptors are expressed in an overall high to low V-D gradient by RGCs. In vitro assays show that Ryk mediates RGC axon repulsion by higher levels of Wnt3, whereas Frizzled receptors mediate an attractant effect of lower Wnt3 levels on dorsal RGC axons (Schmitt et al. 2006).

Mice mutant for the cell adhesion molecules L1 or ALCAM have topographic mapping defects, and L1 has shown to be phosphorylated by EphBs, suggesting that cellular adhesion regulation is also important for D-V mapping (Buhusi et al. 2009, Demyanenko & Maness 2003). Other molecules that may be involved in topographic mapping are a secreted form of the transcription factor engrailed 2 (Brunet et al. 2005), the haywire homolog Phr1, and the type-2 membrane protein Ten_M3 (also called ODZ and teneurin) (Leamey et al. 2007).

Neural Activity Is Required to Generate Fine Structured Topography

Activity-dependent mapping models postulate that neurons with similar activity patterns are more likely to synapse on the same or adjacent target cells than are neurons with different activity patterns (Butts 2002, Debski & Cline 2002). Because topographic mapping in the mammalian visual system occurs before eye opening, this mechanism is thought to rely on spontaneous patterned retinal activity. Throughout the first postnatal week in mice, RGCs discharge highly correlated bursts of action potentials that propagate across the retina as waves; these retinal waves have been theorized to drive synaptic refinement through activity-dependent mechanisms (Butts 2002, Butts et al. 2007, Meister et al. 1991, Wong et al. 1993).

Perhaps the most compelling evidence for this idea came from studies of mice that lack the $\beta 2$ subunit ($\beta 2$ knockout) of the nicotinic acetylcholine receptor (Picciotto et al. 1995, Xu et al. 1999). Retinal waves during the first postnatal week require cholinergic neurotransmission, and $\beta 2$ knockout retinas have defects in the normal patterns of retinal activity (Bansal et al. 2000, Feller et al. 1996, McLaughlin et al. 2003b, Stafford et al. 2009). In these $\beta 2$ knockout mice, multiple aspects of refinement are altered in the dLGN and SC, including retinotopy and eye segregation, demonstrating a correlation between retinal waves as measured in vitro and mapping and behavioral defects in vivo (Cang et al. 2005b, Chandrasekaran et al. 2005, Grubb et al. 2003, McLaughlin et al. 2003b, Mrsic-Flogel et al. 2005, Muir-Robinson et al. 2002, Wang et al. 2009). However, the correlative nature of these studies led to rigorous debates about whether normal retinal waves are required or “instructive” for visual system development (Chalupa 2009, Feller 2009). This uncertainty came about because the in vitro activity patterns of RGCs in $\beta 2$ mutants differed among the labs (Feller 2009, Stafford et al. 2009, Sun et al. 2008), and the in vivo analysis was performed in a global knockout mouse, suggesting that the lack of $\beta 2$ outside the retina could be responsible for the mapping errors.

Xu et al. (2011) recently reintroduced the $\beta 2$ gene into only the RGCs of $\beta 2$ mutant mice and restored many aspects of retinal waves. Although the waves were smaller than those of WT mice, these small waves could rescue topography in monocular regions of the SC. These results suggest that the exact spatiotemporal profile of retinal waves is important for refining topographic connections, but the exact information included in waves, and how they execute refinement, still needs to be determined, especially in vivo. Similarly, the exact molecular mechanisms downstream of retinal waves are not well understood, although cAMP generation via adenylyl cyclase 1 (AC1) is involved (Dhande et al. 2012).

Ephrin-As and Patterned Activity Act Together to Guide Map Formation in the Visual System

The above findings suggest a mechanism whereby EphA/ephrin-A signaling acts to form a rough topography and neural activity acts to cluster correlated inputs to promote its fine structure. In support of this two-step model, intrinsic optical imaging of the ephrin-A2/-A3/-A5 triple mutant mouse finds that their functional SC maps are discontinuous with patches of the SC responding to visual signals from topographically incorrect locations. In other words, large groups of neurons project together to topographically inappropriate locations but can still refine and become integrated into visual circuits. These clusters disappear in mice that are deficient in both ephrin-As and structured neural activity in the retina, leading to a nearly complete loss of topography along the N-T axis of each visual area (Cang et al. 2008a,b; Pfeiffer-Enberger et al. 2006).

Topographic Mapping of Different Retinal Ganglion Cell Types

In the above section, we have treated the retinocollicular projection as a single continuous topographic map that is composed of RGC axons as a single, functionally identical group. In reality, the retinocollicular map is the superimposition of many individual maps that originate from different types of RGCs, each of which is aligned with all the others. Each type of RGC is tuned to a specific visual feature such as motion, color, or changes in luminance. Different RGC types project to specific sets of visual areas and terminate within stereotypic sublaminae within these targets. For example, an ON-OFF direction-selective RGC type projects to the superficial lamina of the SGS, whereas an Off RGC type projects to the deeper SGS (Huberman et al. 2008b, 2009; Kay et al. 2011). Little is known about how different RGC types that project to different sublaminae align with each other. One possibility is that each RGC type sorts topographically using

EphA/ephrin-A signaling and uses a different set of molecules to find the correct lamina. A second possibility is that one RGC type uses EphA/ephrin-A signaling, whereas the other types use activity-dependent (e.g., retinal waves) cues to align with the initial template. Some evidence shows that not all RGCs require

Eph/ephrin signaling to map (Cang et al. 2005a, Pfeifferberger et al. 2006), but we still do not know if the RGCs that map normally in the absence of ephrin-As represent a specific type(s).

ALIGNMENT OF MULTIPLE VISUAL MAPS

Binocular Alignment of Retinotopy

In most mammals, central visual space is seen by both eyes. The extent of this binocular field is determined by eye position and varies across species (Leamey et al. 2008). In primates and humans, RGCs in the temporal retina that view the contralateral half of the binocular visual field project to the SC and dLGN in the same side (**Figure 4**), where they meet the axons of the nasal RGCs from the other eye that see the same hemifield. Although mice have a smaller binocular field because their eyes are positioned more laterally, similar projection patterns are seen except that there is a much smaller percentage of ipsilaterally projecting RGCs (~3–5%, Petros et al. 2008) that originate from the ventral-temporal crescent of the retina. The crossed and uncrossed axons are segregated into eye-specific layers in the SC and dLGN. Within each layer, the inputs are organized retinotopically, and the two maps are aligned between the layers. Relay neurons from different layers of the dLGN project to V1, where they converge onto the same postsynaptic neurons, resulting in the first stage of binocular processing.

For proper binocular alignment in the brain, individual RGC axons have to determine correctly whether to cross the midline. This decision is facilitated by the interaction between molecules expressed on the RGC growth cones and those at the optic chiasm (for a review, see Petros et al. 2008). Once the crossed and uncrossed axons reach their primary targets, their retinotopic maps align. This alignment requires the topography of ipsilateral axons to be reversed along the N-T axis relative to the contralateral axons. In the SC, for example, the temporal RGC axons of the contralateral

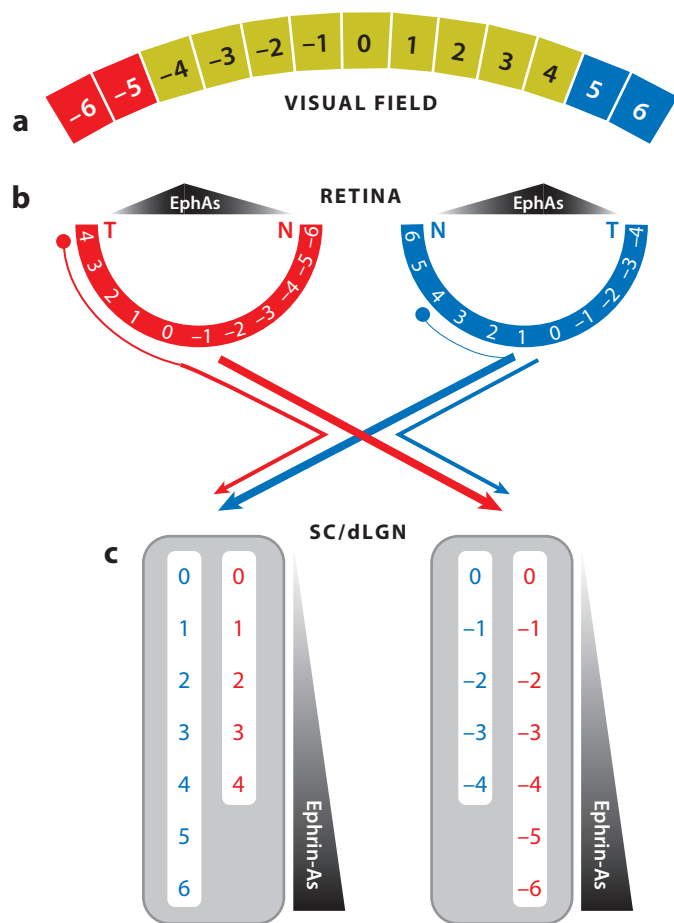


Figure 4

Binocular alignment of retinotopy. The visual field has both monocular and binocular areas (represented here as regions from -6 to 6, *panel a*) that project onto the retina of each eye (*panel b*); the central part is projected onto both eyes. The retinal projections of each eye are segregated in the SC/dLGN and are topographically aligned (*panel c*). The alignment requires that RGCs of different N-T coordinates of each eye project to nearby areas. For example, position 4 in the left SC/LGN receives inputs that originate from the most temporal retina of the left eye (red) and from the nasal retina of the right eye (blue). A center-to-periphery gradient of EphAs in the retina gives these axons similar levels of mapping labels to align on the basis of their similar sensitivities to a gradient of ephrin-As in the target.

retina terminate more anteriorly and nasal axons more posteriorly, and the order is reversed for axons from the ipsilateral retina (**Figure 4**). How can the same set of guidance cues, such as ephrin-As/EphAs, mediate the formation of two maps that have opposite polarity? Sperry predicted that, in binocular species, the N-T gradient of the mapping labels would become a radial gradient (center to periphery) to give nasal axons from one eye and temporal axons from the other eye similar levels of mapping labels (Sperry 1963). Remarkably, EphA receptors are indeed expressed in a center high and peripheral low gradient along the N-T axis in the developing retinas of both humans and ferrets (Huberman et al. 2005, Lambot et al. 2005), consistent with their potential roles in binocular retinotopic mapping.

A similar center-to-peripheral gradient is more difficult to demonstrate in mice because only a very small number of RGCs project ipsilaterally in this species. Although not noted in the original studies, such a gradient may indeed exist in mice (Feldheim et al. 2000, figure 6c), suggesting a remarkable conservation in evolution. Consistent with their expression patterns, EphA/ephrin-A signaling may indeed be involved in aligning the contralateral and ipsilateral maps in mice. In ephrin-A2/A5 double knockout mice, the binocular alignment of functional maps in the SC is disrupted (Haustead et al. 2008), though the interpretation is complicated because these mice have mapping errors in both contralateral and ipsilateral projections (Feldheim et al. 2000, Haustead et al. 2008). In contrast, a transmembrane protein, Ten_m3, is required only for mapping the ipsilateral projections to the dLGN (Leamey et al. 2007). The exact mechanism of this process is still unknown, as Ten_m3 expression is not restricted to the ipsilaterally projecting RGCs (Leamey et al. 2007). The binocular misalignment is associated with visual behavioral deficits in both ephrin-A2/A5 knockouts and Ten_m3 knockouts, and the behavior deficits could be reversed by occluding or silencing one of the two eyes (Haustead et al. 2008, Leamey et al. 2007).

In the developing dLGN and SC, the RGC axons from the two eyes are initially intermingled and then retract to segregate into eye-specific layers (Godement et al. 1984, Jaubert-Miazza et al. 2005, Jeffery 1985, Lund et al. 1974). These data suggest that interactions between the crossed and uncrossed retinal axons may be important for binocular alignment. Indeed, when the contralateral projection is removed by neonatal enucleation in rats, the ipsilateral projections from the remaining eye fail to establish normal topography along the N-T axis in the SC (but not in the dLGN; Reese 1986), suggesting that the ipsilateral projections may use the contralateral ones as a template to align the two maps. The interactions between the eye-specific axons may be mediated by molecular cues, such as EphAs/ephrin-As (Haustead et al. 2008) and/or spontaneous neuronal activity. Xu et al. generated a line of transgenic mice that express the $\beta 2$ subunit of the acetylcholine receptor only in the RGCs (see above) and showed that they have smaller retinal waves during development. Retinotopic refinement in these mice is normal except in the binocular zone of the dLGN and SC (Xu et al. 2011), suggesting that the activity-dependent segregation of contralateral and ipsilateral axons is mechanistically linked with retinotopic refinement and potentially with their alignment as well.

The first stage of binocular visual processing takes place in V1. The converging eye-specific geniculocortical inputs are aligned retinotopically, giving rise to binocular neurons that represent similar points in the visual space through both eyes. The development of cortical retinotopy is understood better for the azimuth map through the contralateral eye, which is guided by a combination of EphA/ephrin-A signaling and spontaneous neural activity driven by retinal waves (Cang et al. 2005a,b, 2008a). In mice, the retinotopic order of geniculocortical projections appears before P8 when RGC axons from the two eyes are still largely intermingled in the dLGN (Pfeiffenberger et al. 2005). During this period, graded EphA expression is seen along the M-L axis of V1 (Cang et al.

Orientation selectivity: the property that visual cortical neurons are more responsive to bars presented over a narrow range of orientation

Orientation maps: cortical organization where neurons within a vertical column prefer similar orientations

2005a), and ephrin-As display a single gradient in the dLGN along the lines of retinotopy regardless of eye-specific layers (Feldheim et al. 1998). Consequently, the same developmental processes that guide single map formation will likely map both inputs together to achieve binocular alignment. Consistent with this notion, the binocular misalignment seen in the dLGN of *Ten_m3* knockout mice propagates to V1 and results in binocular cells that have mismatched receptive fields through the two eyes (Leamey et al. 2007, Merlin et al. 2012).

Binocular Matching of Orientation Tuning

Two major transformations occur when visual information reaches the cortex. First, cortical cells are binocular as discussed above; and second, cortical cells are selective for stimulus orientation (Hubel & Wiesel 1962). As a result, binocular cells in the cortex are tuned to similar orientations through the two eyes (Bridge & Cumming 2001, Ferster 1981, Hubel & Wiesel 1962, Nelson et al. 1977, Wang et al. 2010), a property presumably necessary for normal binocular vision. In carnivores and primates, cortical neurons with similar orientation preferences are organized into columns (Blasdel & Salama 1986, Grinvald et al. 1986, Hubel & Wiesel 1962), and the maps of orientation columns are also binocularly matched (Crair et al. 1998). According to the feedforward model proposed originally by Hubel & Wiesel (1962), orientation selectivity arises from the specific arrangement of geniculate inputs, and the preferred orientation of individual cortical cells is determined by the layout of the elongated ON (responding to light increment) and OFF (responding to light decrement) subregions in their receptive fields (Ferster & Miller 2000, Priebe & Ferster 2012). An alternative series of models, the feedback models, propose that orientation selectivity is an emergent property of intracortical circuitry (Adorjan et al. 1999, Ben-Yishai et al. 1997, Somers et al. 1995). Although researchers are still debating the exact synaptic mechanisms of

orientation selectivity, many studies have investigated its development and, more recently, its binocular matching.

Orientation selectivity develops after retinotopic mapping and eye-specific segregation in the dLGN and matures after eye opening (Espinosa & Stryker 2012, Huberman et al. 2008a). The development of monocular orientation selectivity does not require visual experience, but its binocular matching, at least in mice, does. Mice do not have orientation columns, but individual V1 neurons are highly selective (Niell & Stryker 2008) and prefer similar orientations through the two eyes (Wang et al. 2010, Sarnaik et al. 2013). In young mice (P20, ~1 week after eye opening), orientation tuning is binocularly mismatched, and the mismatch decreases and reaches adult levels by P30. The binocular matching process occurs in a time window that coincides with the critical period for ocular dominance plasticity (Espinosa & Stryker 2012) and requires normal binocular vision (Wang et al. 2010).

In contrast, the matching of large-scale orientation maps appears to be experience-independent. Binocularly matched orientation maps are seen in very young kittens (Crair et al. 1998) and can develop even in the absence of coordinated binocular activity (Godecke & Bonhoeffer 1996). The degree of binocular orientation matching at the single-cell level may improve further after matching the maps in an experience-dependent manner, but this possibility has not been studied in species that have orientation maps. One study reported a small number of binocularly mismatched V1 neurons in visually deprived kittens (Blakemore & Van Sluyters 1974, figure 4), consistent with the results in mice (Wang et al. 2010).

Development of orientation selectivity requires spontaneous neuronal activity in the visual cortex (reviewed in Espinosa & Stryker 2012, Huberman et al. 2008a). In ferrets, binocularly correlated spontaneous activity exists in the LGN before eye opening because of corticogeniculate feedback (Weliky & Katz 1999). Whether such activity contributes to binocular matching of orientation is not known, but

a Hebbian plasticity-based model predicts that binocularly matched orientation maps would develop if the same-polarity dLGN inputs from the two eyes (On-On, Off-Off) are temporally correlated at the same retinotopic locations (Erwin & Miller 1998). Finally, the experience-independent binocular matching of orientation maps suggests that molecular cues may be involved in the development of orientation selectivity, orientation maps, and their binocular matching. However, such molecules have not yet been discovered.

Template-Driven Alignment between Retinal and Cortical Visual Maps in the Superior Colliculus

In addition to the retinal inputs from both eyes, the SC also receives cortical projections that are aligned with the retinocollicular map (Rhoades & Chalupa 1978). This corticocollicular projection provides a link between the two major streams of visual processing: the retinocollicular pathway, used for some reflexive visual behaviors; and the retino-geniculo-cortical pathway involved in conscious vision.

Two distinct models can explain how the retinocollicular and corticocollicular maps become aligned in the SC. A gradient-matching model postulates that gradients of molecules expressed by both V1 and RGC axons match graded labels expressed in the SC to specify each map (**Figure 5a**). In this case, corticocollicular mapping is independent of retinocollicular mapping, but because both projections use information provided by the same target molecules, they become aligned. A retinal-matching model for map alignment proposes that Hebbian-type mechanisms or direct axon-axon interactions are used to direct V1 and RGC axons that monitor the same point in space to terminate in the same region of the SC (**Figure 5b**). In this model, the retinocollicular map is established using the mapping mechanisms described above. V1 axons would then grow in and form synapses with the SC neurons that are innervated by the RGCs that share common activity patterns or cell surface

molecules. In this mechanism, the retinal axons serve as a template that orients and organizes the alignment with incoming projections.

The *Islet2-EphA3* knockin ($EphA3^{ki/ki}$) mouse described above was an ideal genetic tool to distinguish between these models. Using optical imaging of intrinsic signals to visualize functional maps in the brain, Triplett et al. (2009) found that the $EphA3^{ki/ki}$ mouse has a duplicated functional retinocollicular map but has a single functional map in V1 (**Figure 5c**). Because exogenous EphA3 expression is limited to RGCs, the gradient-matching model predicts that the V1-SC projections remain normal, thus resulting in a misalignment of the retinocollicular and corticocollicular maps. In contrast, the retinal-matching model predicts that cortical projections in the SC will align with the altered retinal projections. Triplett et al. found that the corticocollicular projections indeed bifurcate to align with the duplicated retinocollicular maps in $EphA3^{ki/ki}$ mice, supporting the retinal-matching model (**Figure 5c**). Furthermore, disruption of the spontaneous cholinergic retinal waves in $EphA3^{ki/ki}$ mice prevented map alignment. Taken together, these data showed that the V1 projection aligns with a preexisting retinocollicular map by matching activity patterns that propagate throughout the visual system prior to eye opening (Triplett et al. 2009). EphA/ephrin-A signaling and/or other guidance cues are likely used to specify the overall polarity and perhaps rough topography of the V1-SC map, as the V1 axons of enucleated mice and *Math5* mutants still project to the proper half of the SC in the absence of retinal inputs (Triplett et al. 2009).

While spontaneous retinal activity drives the topographic development of V1 inputs, visually evoked activity further consolidates these inputs to form functional circuits. Eye opening induces rapid local branching of cortical axons in the SC and causes an increase in the number and strength of the synapses that individual dendrites oriented vertically (DOV) neurons receive. Visual deprivation disrupted these changes and reduced the number of

Hebbian plasticity:

form of synaptic plasticity in which synaptic efficacy increases as a result of correlated activity between pre- and postsynaptic cells

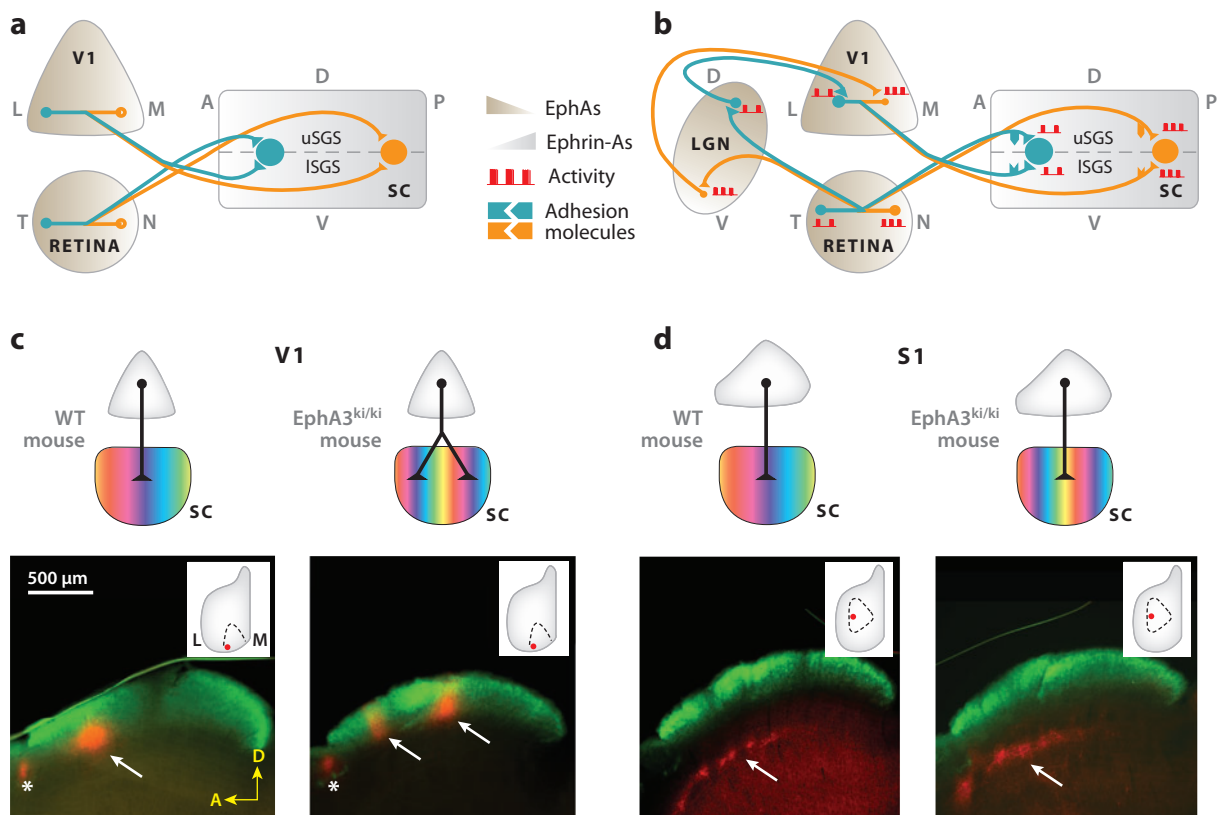


Figure 5

Alignment of the primary visual cortex (V1) and the primary somatosensory cortex (S1) inputs with retinocollicular maps in the superior colliculus (SC). (*a–b*) Models of visual map alignment in the SC. In the gradient-matching model (*a*), graded expression of EphA receptors in both the retina and V1 are used to guide topographic mapping in the SC, which expresses repulsive ephrin-A ligands in a gradient in both recipient layers. In the retinal-matching model (*b*), retinocollicular mapping is established first through the use of graded EphAs and ephrin-As. V1 projections then terminate in areas with similar activity patterns or with RGCs expressing complementary cell surface molecules. Abbreviations: N, nasal; T, temporal; A, anterior; P, posterior; D, dorsal; V, ventral; L, lateral; M, medial; uSGS, upper stratum griseum superficiale; ISGS, lower stratum griseum superficiale; WT, wild type. (*c–d*) Different outcomes of V1 and S1 inputs to the SC in EphA3^{ki/ki} mice. (*c*) An injection of Dil into central V1 traces axons that project to the central SC (white arrow). A similar injection bifurcates and projects to each map in the EphA3^{ki/ki} mouse (white arrows), showing that the retinal inputs to the SC direct V1 inputs; *, pretectal nucleus. (*d*) S1 axons do not alter their projection patterns in the EphA3^{ki/ki} mouse, resulting in misalignment between retinal and SC maps. Diagrams and data are from Triplett et al. 2009, 2012.

spines on the V1-recipient dendrites of the DOV neurons (Phillips et al. 2011), suggesting a role for experience-driven activity in the final stages of map alignment.

MULTIMODAL MAP ALIGNMENT

Even though the above map-alignment examples are all from the visual system, the brain

also integrates maps from different modalities. In the SC, aligning sensory and motor maps allows proper integration of sensory information and initiation of orienting movements to redirect attention toward the stimulus. Studies have shown that both activity-dependent and activity-independent mechanisms contribute to the establishment of map alignment between modalities.

Aligning Visual and Somatosensory Maps in the Superior Colliculus

Neurons in S1 project axons topographically to an SC lamina deeper than do V1 axons and provide a link between touch and vision. However, unlike V1 axons, S1 neurons do not receive any retinal input and do not change in the EphA3^{ki/ki} mouse (**Figure 5d**), confirming that retinal input into the SC does not influence the S1–SC map. Instead, the S1–SC topography requires ephrin-As because the ephrin-A2/A3/A5 knockout mouse has a completely disrupted S1–SC projection (Triplett et al. 2012). This observation suggests that the S1–SC projection matches EphA/ephrin-A gradients to map topographically. In combination with the above studies of V1–SC map development, a model has emerged in which distinct afferents use different mechanisms to achieve alignment.

Aligning Visual and Auditory Maps

Unlike in the visual system, where space naturally projects onto the photoreceptors across the retina, the topographic representation of the auditory space must be computed in the brain (Knudsen et al. 1987). The location of a sound source is determined by using spatial cues arising from the interactions of the head and ears with the sound stimulus, such as interaural timing differences (ITDs) and interaural level differences (ILDs). These cues are processed in frequency-specific channels in several brain stem nuclei, then combined in the posterior midbrain to generate a map of auditory space (Knudsen et al. 1987). The anatomy and computation of sound localization and its development have been studied in several species, and most notably in barn owls, where the map of auditory space is first seen in the external nucleus of the inferior colliculus (ICX) (Knudsen & Konishi 1978b). The ICX combines inputs across frequency-specific channels arriving from the central nucleus of the inferior colliculus (ICC), where ITDs and ILDs are still organized according to sound frequency (tonotopically). The auditory maps

in the ICX (and brachium of the inferior colliculus in mammals; King 1999) are conveyed to the SC/OT and topographically align with maps of the visual space and of the body surface (Knudsen & Brainard 1995). As a result, many neurons in the intermediate and deep layers of the SC are bimodal or trimodal and display closely aligned auditory and visual receptive fields in space (Drager & Hubel 1975, King 1999, King & Hutchings 1987, Knudsen 1982, Knudsen & Brainard 1995, Middlebrooks & Knudsen 1984).

Compared with the retinotopic map, the map of auditory space in the SC develops later and requires sensory experience. This finding is not surprising given the extensive growth and changes in head size and ear position during early life and their individual differences, which alter the values of ITD and ILD cues important for sound localization. The maturation of auditory spatial tuning and space map in the SC is a protracted process that takes place after hearing and vision onset (Vachon-Preseau et al. 2009, Wallace & Stein 1997, Withington-Wray et al. 1990a). In animals that were raised without location-specific acoustic cues or those that were visually deprived, auditory neurons in the SC/OT display disrupted spatial tuning and the auditory map is misaligned with the visual map (Efrati & Gutfreund 2011; King & Carlile 1993; Knudsen et al. 1991; Withington-Wray et al. 1990a,b). Some aspects of the SC auditory maps in the blind-reared animals are normal, suggesting that innate genetic programs may establish a crude map of space (Knudsen & Brainard 1995), although the molecules involved have not been identified.

How visual experience aligns the auditory and visual maps during development has been studied with manipulations that cause the two maps to misalign (King 1999, Knudsen 2002). For example, in barn owls, wearing prismatic spectacles that optically displace the visual field causes a misalignment between the visual and auditory spaces. During a sensitive period in early life, prism-reared owls can learn the new associations between auditory spatial cues and the shifted visual world, and the auditory and

Interaural timing differences (ITDs):

the difference in the timing of a sound reaching each ear

Interaural level differences (ILDs):

the difference in the loudness of the sound entering each ear

visual maps in the OT are once again in register (Brainard & Knudsen 1998, Knudsen et al. 1991). In this process, the retinotectal map does not change, but the auditory receptive fields of individual neurons and the map of the auditory space in the OT shift to match the visual displacement. As a result, turning toward the sound source again brings it to the line of sight through the prisms. Such adaptive changes are also seen in juvenile ferrets, in which surgical deviation of one eye results in a corresponding shift of the auditory space map in the contralateral SC (King et al. 1988).

Knudsen (2002) identified the ICX as the site of plasticity in barn owls, in which the ICC-to-ICX projections became broader and the spatial tuning of ICX neurons was altered adaptively in the prism-reared animals (Knudsen 2002). Because auditory map development and plasticity are under the instruction of visual experience, visual inputs must reach the ICX. The instructive visual signal originates from the OT and is topographic (Hyde & Knudsen 2001), as a small lesion in the OT eliminates auditory map plasticity only in the corresponding region in the ICX (Hyde & Knudsen 2002; see also King et al. 1998 for similar experiments in ferrets). Visually evoked responses are indeed seen in the ICX of anesthetized owls, but only after GABAergic transmission is blocked in the OT, indicating that the instructive signal from the OT to the ICX is gated by inhibition (Gutfreund et al. 2002). These visual inputs have spatially restricted receptive fields that align with the auditory receptive fields of the same sites in normal owls (Gutfreund et al. 2002). Together, these experiments support the idea that the visual map in the SC/OT is used as a topographic template to guide the transformation of spatial cues into the auditory space map, thereby achieving visual and auditory map alignment.

Aligning Visual and Motor Maps

The SC is well known for its role in initiating orienting behaviors, and the motor commands are sent out by neurons in the deep layers.

These neurons project to several premotor and motor nuclei that control muscles for eye, head, ear, whisker, and body movements (Gandhi & Katnani 2011). The topographic organization of these motor functions is best studied for saccadic eye and head movements (Wurtz & Albano 1980). In primates, the motor map in the deep layers of the SC is in register with the retinotopic map in the superficial layers, such that the direction of gaze is shifted to the site of the visual stimulus after the evoked saccade (Schiller & Stryker 1972, Sparks et al. 1990, Wurtz & Goldberg 1972). In barn owls, which have very limited eye movements, the maps of head movement are similarly aligned with the visual and auditory maps in the OT (du Lac & Knudsen 1990), enabling proper visual/auditory-orienting behavior.

Both experience-dependent and -independent processes are likely involved in aligning sensory and motor maps during development. Electrical stimulation of the SC can produce eye movements in neonatal kittens, and the evoked movements show some features of topographic organization even before visual stimuli can activate neurons in the SC (Stein 1984, Stein et al. 1980). Barn owls that were reared with closed eyelids could still develop a head movement map with correct global polarity, but the movement map displayed substantial misalignment with both the visual and the auditory maps (du Lac & Knudsen 1991). Whether and how visual experience is required for visual-motor map alignment in mammals remain unknown. The lack of progress is perhaps because the mouse, a great model in developmental studies as evidenced in this review, was assumed to lack eye movements because mice do not have a fovea. However, recent studies have shown that mice do, in fact, have saccade-like rapid eye movements that can be both spontaneous and evoked by SC electrical stimulation (Sakatani & Isa 2007, 2008). The amplitude and direction of the evoked eye movements appear to be organized in a map that has the same global polarity as the retinotopic map in the SC (Wang et al. 2011). Given the powerful genetic

tools and available transgenic lines that alter the visual map (e.g., the EphA3^{ki/ki} mouse), mice will become a useful model to reveal the molecular and activity-dependent mechanisms of motor map development and sensorimotor alignment.

CONCLUSIONS AND FUTURE DIRECTIONS

Much progress has been made to understand how topographic maps are established; these data have provided important foundations for studying how multiple maps are aligned during development. Three main factors in map alignment have been revealed: molecular guidance cues, activity-dependent processes driven by patterned spontaneous activity, and experience-driven developmental plasticity. Although all three mechanisms are required in map alignment, their relevant contributions differ and depend on the logic of the particular alignment in question. For example, as EphA/ephrin-A gradients map both retinotopic projections in the SC and somatotopic projections in S1, the same sets of gradients can naturally be used to guide S1 inputs to the SC to align with visual maps. A similar process is probably used to map V1 projections to the SC, but because retinal and V1 inputs share patterned spontaneous activity, their alignment can be further driven by a template-matching process instructed by retinal waves. However, to align auditory space maps and eye-movement motor maps with the visual map, molecular gradients and spontaneous activity are not enough. This is because simple Eph/ephrin gradients seen in the developing visual system cannot guide the formation of the auditory space maps and eye-movement maps, where the represented values are computed and transformed from their inputs. Furthermore, these systems and the visual system do not share spontaneous activity patterns that carry spatial information. As a result, for these maps to align, experience is needed to provide the necessary instructive signal from one system to the other. The instructive signal originates from the visual system in both visual-

auditory and visual-motor alignment, presumably because vision provides the most accurate and reliable information and the main purpose of the alignment is to shift the line of sight.

Despite these advances, studies of map alignment are still very descriptive. A major goal now is to reveal the molecular and synaptic mechanisms used to mediate map formation and alignment during development and to understand how they work together. First, clearly much still needs to be learned about other, non-Eph/ephrin molecules that must be involved in establishing computation maps, such as maps of orientation columns in the visual cortex of carnivores and primates, maps of auditory space in the SC/OT, and eye/head movement maps in the SC/OT. New genomic technologies such as RNA sequencing, which allow researchers to compare transcriptomes of different cell types or to compare the same cell types under different conditions (e.g., dark-reared versus normal), can reveal genes that are involved in these processes in traditionally nongenetic model organisms.

Second, what are the patterns of spontaneous activity in vivo when the maps are being formed and aligned, not only in the retina but in all visual areas? This unknown can now be addressed with recent technical advances in multiphoton imaging, which allows investigators to monitor neural activity at single-cell resolution of a large number of cells (Siegel et al. 2012, Zhang et al. 2012). Indeed, the recent report of retinal waves imaged in vivo is an exciting development along this direction (Ackman et al. 2012). Future experiments that combine these imaging techniques with transgenic mice and optogenetics will provide important information about how retinal waves instruct topographic map formation and alignment.

Furthermore, what are the synaptic and circuit mechanisms for activity-/experience-dependent map formation and alignment? Hebbian plasticity seems to be the most likely candidate, but direct evidence for this hypothesis is still lacking. It will be critical to follow connectivity patterns and functional properties of individual neurons during map formation

rather than studying only the consequences of activity manipulation and visual deprivation after development.

Finally, a long-term goal of this field of research is to understand the behavioral consequences of animals that have defects in multimodal processing. These studies will generate valuable insights to help us understand neurodevelopmental diseases such as autism spectrum disorders, psychosis, and mental retardation, each of which displays impairments in aspects of sensory integration, sensorimotor

transformation, social interactions, and language development. The normal development of these skills requires proper information association and integration in the nervous system, which often occur in an experience-dependent manner during critical periods in early life. By studying map alignment, the structural substrate of multichannel integration, and how it is instructed by sensory and motor experiences, we may better understand the pathogenic mechanisms of these disorders and reveal new interventions to treat them.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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